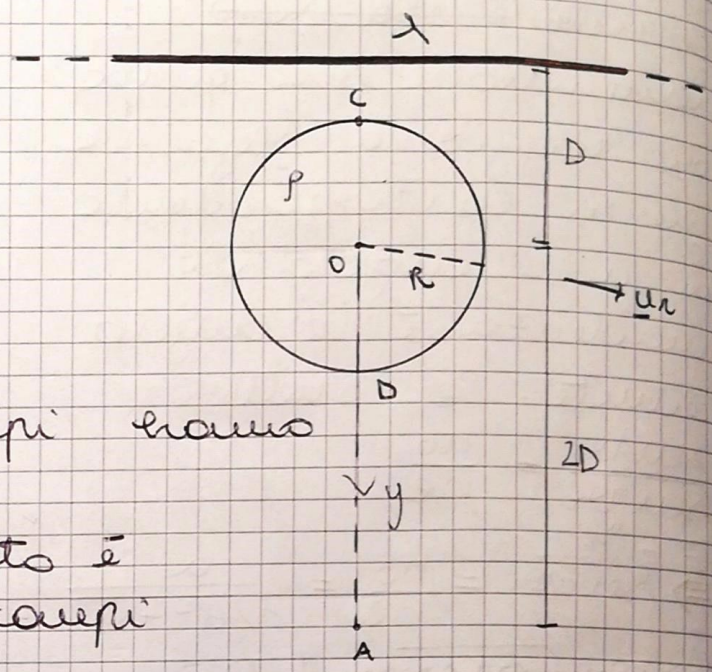
APPELLO 10/07/2018

ESERCIZIO 1

$$\underline{E} = \frac{\rho}{3\varepsilon_0} r \, \underline{u_r} \qquad r \leq R$$

$$\underline{E} = \frac{\rho R^3}{3\varepsilon_0 r^2} \underline{u_r} \qquad r > R$$



Il campo generato dal filo è

$$\underline{E}' = \frac{\lambda}{2\pi\varepsilon_0 r'} \underline{u_r}'$$

Ora lungo y i due campi hanno la stessa direzione.

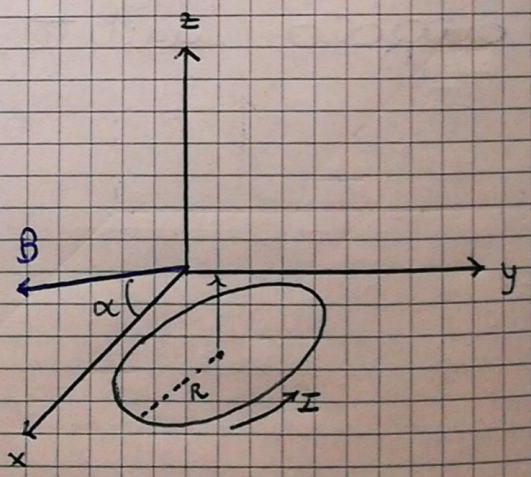
Il campo totale nel punto è dato dalla somma dei campi

$$\underline{E}_{TOT}(A) = \left(\frac{\rho R^3}{3\varepsilon_0 4D^2} + \frac{\lambda}{2\pi\varepsilon_0 3D} \right) \underline{u_y}$$

$$\Delta V = V_C - V_B = -\int_B^C E_{TOT} \, dr =$$

$$= + \int_{-R}^{R} \frac{\rho}{3\varepsilon_0} y + \frac{\lambda}{2\pi\varepsilon_0 (y+D)} dy = \frac{\lambda}{2\pi\varepsilon_0} \ln\left(\frac{R+D}{D-R}\right) > 0$$

ESERCIZIO 2



a) $\underline{m} = I\pi R^2 \underline{u_z}$

b) $\underline{M} = \underline{m} \times \underline{B} = mB \operatorname{sen}\left(\frac{\pi}{2} - \alpha\right) \underline{u_y} =$

$= I\pi R^2 B \cos\alpha \, \underline{u_y}$

c) $L = -\Delta U_m = -L^{EST}$

$U_m^{(i)} = -I\pi R^2 B \operatorname{sen}\alpha$

$U_m^{(f)} = -I\pi R^2 B \cos\left(\frac{\pi}{2} + \alpha\right) =$

$= I\pi R^2 B \operatorname{sen}\alpha$

$L^{EST} = \Delta U = 2\pi I R^2 B \operatorname{sen}\alpha$